LINK TO TINKERCARD :

https://www.tinkercad.com/things/i2I4QSWnTir/editel?returnTo=%2Fdashboard&sharecode=f3UN2oqoR-qSg0fx49aLxLnmgj3lwqxlPcXocQyGc6c

```
const int tempPin = A0;
const int ledPin = 8;
const int motorPin = 9;

void setup() {
  pinMode(ledPin, OUTPUT);
  pinMode(motorPin, OUTPUT);

  Serial.begin(9600);
  Serial.println("CPS Monitoring and Control System");
  Serial.println("------------------------------");
}

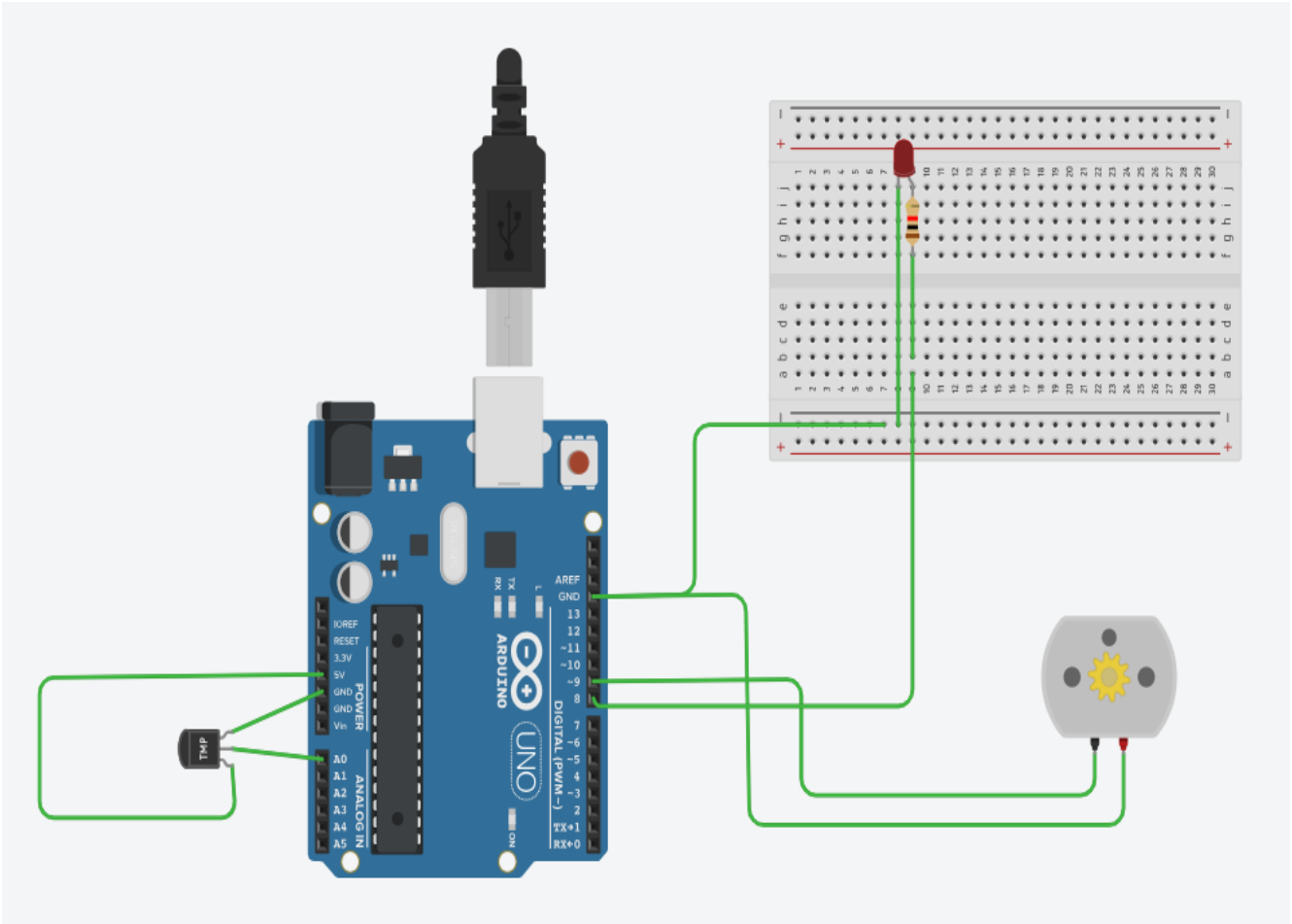
void loop() {
  int sensorValue = analogRead(tempPin);

  float voltage = sensorValue * (5.0 / 1023.0);
  float temperature = (voltage - 0.5) * 100.0;

  Serial.print("Temperature: ");
  Serial.print(temperature);
  Serial.println(" C");

  if (temperature >= 30.0) {
    digitalWrite(ledPin, HIGH);
    digitalWrite(motorPin, HIGH);
    Serial.println("High Temperature - Motor ON");
  } else {
    digitalWrite(ledPin, LOW);
    digitalWrite(motorPin, LOW);
    Serial.println("Normal Temperature - Motor OFF");
  }

  Serial.println("------------------------------");
  delay(1000);
}
```


IOREF
RESET
3.3V
5V
GND
GND
Vin
POWER
A0
A1
A2
A3
A4
A5
ANALOG IN
TMP
ARDUINO
UNO
AREF
GND
13
12
~11
~10
~9
8
7
~6
~5
4
~3
2
TX→1
RX←0
DIGITAL (PWM~)